

WAI YAN MIN MIN (ZACK)

Full Stack AI Engineer | RAG, Agents & Production LLM Systems

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LinkedIn | GitHub | Portfolio | Stack Overflow

Full Stack AI Engineer who owns features from idea to production — React/TypeScript frontends, Python/Node.js APIs, and LLM systems including RAG pipelines, prompt refinement, and agent workflows with tool calling and automation. Ships reliable AI features on AWS with a focus on quality, latency, and practical business impact.

CORE SKILLS

AI & Orchestration: LLMs, LangChain, RAG, AI Agents, Tool Calling, Prompt Engineering, Embeddings, NLP, Workflow Automation, TensorFlow, PyTorch, scikit-learn

Full Stack: React, TypeScript, Tailwind CSS, Next.js, Node.js, FastAPI, Python APIs, RESTful APIs, GraphQL, React Native, Flutter

Programming: Python, JavaScript, TypeScript, Node.js, SQL, Java, C++, C#, PHP

Data & Retrieval: PostgreSQL, MySQL, MongoDB, DynamoDB, SQLite, Prisma, embeddings/vector-ready data modeling for RAG

Cloud & DevOps: AWS (EC2, S3, RDS, CloudFront, Route 53), Docker, CI/CD, GitHub Actions, Vercel, Railway, production monitoring & troubleshooting

Integrations & Tools: n8n, third-party APIs, messaging/email automation, Git/GitHub/Bitbucket, Streamlit, Jupyter

Delivery: end-to-end feature ownership, rapid iteration with product stakeholders, Clean Architecture, scalable API design

PROFESSIONAL EXPERIENCE

Freelance Developer | Singapore

February 2026 – Present | Remote

Owned customer-facing web features end-to-end for corporate event platforms (500+ attendees), from concept through production release.

Built responsive frontends and backend APIs with secure auth, reliable data capture, and iteration based on client feedback.

Deployed and operated systems on AWS, improving reliability, security, and availability under peak live traffic.

Integrated hosting, DNS, and third-party services so client sites stayed live through high-demand registration windows.

Delivered production systems for clients including Infineon, UOB, and Siemens.

Flutter Developer | Timeworld (Singapore)

December 2024 – November 2025 | Remote

Owned the full delivery lifecycle as the sole full-time developer — build, test, release, and iterate production mobile features.

Implemented Bloc architecture and clean state management for maintainable, scalable cross-platform releases.

Built deep-linking and release-ready features for production app distribution across platforms.

Optimized performance and localized the app across five languages for global user reach.

Partnered on release readiness, production bug fixes, and stable updates for end users.

Full Stack Developer | Pixora Lab

June 2024 – January 2025 | Remote

Developed full-stack web applications across React-style frontends, backend APIs, and databases with rapid feature delivery.

Designed scalable RESTful APIs and connected third-party services to power product workflows.

Deployed and maintained applications on AWS (EC2, S3, RDS, CloudFront, Route 53).

Implemented CI/CD, monitored production health, and troubleshoot incidents to improve reliability.

Collaborated with cross-functional teams to ship secure, scalable features quickly.

Frontend Developer | Mom&Pop (Thailand)

December 2023 – February 2024 | Remote

Collaborated with a five-person team to ship customer and shop-owner mobile apps with responsive, production-ready UX.

Integrated IoT peripherals (printers, Bluetooth printers, scanners), enabling higher real-world adoption in retail settings.

Partnered with the field team to diagnose device-specific production bugs and ship stable fixes.

Freelance Developer | Upwork

June 2023 – June 2024

Delivered a full-stack project from scratch for a US-based client — setup, feature development, and handoff.

Implemented Provider-based state management and integrated backend APIs for reliable client-server data flow.

Developed backend components in C# and supported end-to-end feature delivery across the stack.

EDUCATION

Bachelor's Degree in Data Science | University of Hertfordshire, Singapore

January 2025 – April 2026

Graduated with First Class Honors, GPA 3.98 out of 4.5

Diploma in Info Communication | PSB Academy, Singapore

April 2024 – December 2025

Covered Python, OOP, networking fundamentals, and cybersecurity basics.
Awarded Top Student out of 80 students in the class.

Diploma in Computing I NCC Education (UK)

January 2022 – August 2024

Graduated with GPA 4.0 out of 5.0.

Built a PHP website as the final project; founded and managed the school football club.

PROJECTS

Cross-Platform Ecommerce Web App with Admin Panel (iOS, Android, Web)

React Native, Convex, Realtime Database | June 2026 – Present

Building a cross-platform ecommerce app (iOS, Android, Web) with React Native and Convex realtime backend/database.

Implementing core shop flows and messaging between shoppers and clothing stores via product APIs and realtime data.

Developing brand-owner admin dashboards for order tracking, analytics, and standard ecommerce operations.

AI Costudy Pilot (Full-Stack AI Web App)

React/Next.js, TypeScript, Python APIs, PostgreSQL, Prisma, AWS RDS, S3 | RAG-style PDF Q&A, summaries, quizzes

Designed and built a customer-facing AI study product end-to-end: document AI summaries, semantic Q&A, and quiz generation.

Implemented document-backed retrieval workflows over user content with access-controlled data on AWS RDS and S3.

Shipped secure auth, profile management, and API key controls to improve reliability and safe AI usage in production.

Deployed frontend and backend services; resolved CORS, session, and build issues while iterating toward stable releases.

GitHub Issue Analysis & AI Agent Workflow Automation

n8n, LangChain, LLMs, Tool Calling, Docker, Railway | multi-step AI workflows

Built an AI agent workflow that analyses GitHub issues with LangChain prompts, then orchestrates multi-step automation.

Connected LLM outputs to messaging/email notifications and third-party APIs for real-world triage handoff.

Containerized and deployed the pipeline; reduced manual engineering triage time through reliable agent-style automation.

AI Resume Screening & Ranking (HR)

Python, FastAPI, Embeddings, Semantic Similarity, Flutter

Built a production-style AI screening service with FastAPI: parse resumes, score job fit, and return ranked shortlists.

Used embedding-based semantic matching (beyond keywords) and explainable signals to improve ranking consistency and quality.

Delivered end-to-end API + UI for HR evaluation workflows with practical latency and reliability considerations.

Computer Vision Models Performance Research

VGGFace2, CNN, ResNet50 | Singapore | December 2025 – April 2026

Evaluated and compared computer vision models under varied parameters and sample sizes on the VGGFace2 dataset.

Trained a baseline CNN and benchmarked against ResNet50 transfer learning to assess model performance trade-offs.